

Retro Foresight — Signal Cards

N1

THE GLOBE AND MAIL

JULY 14, 2000

SARAH HENNESSEY, HEALTH CORRESPONDENT

FDA Clears First Robotic System for General Surgery

WASHINGTON — The U.S. Food and Drug Administration this week granted clearance to the da Vinci Surgical System for use in general laparoscopic surgery, marking the first time a fully integrated robotic platform has been approved for broad surgical use. Developed by Intuitive Surgical Inc. of California, the machine allows a surgeon to operate from a seated console, manipulating robotic arms through incisions smaller than a fingertip.

Supporters argue the system is a fundamental shift in what's possible in the OR. The robotic arms filter out hand tremors, scale down the surgeon's movements to microscopic precision, and provide a magnified 3D view of the surgical field. "This is not automation," said an early adopter at Johns Hopkins. "The surgeon is still entirely in control; the robot just makes their hands better."

Critics, however, are unconvinced. Several surgeons contacted for this article said experienced laparoscopic surgeons already achieve excellent outcomes without robotic assistance. "The best surgery in the world won't mean much if only three hospitals in the country can afford the machine," said a senior surgeon at a Toronto teaching hospital. Independent analysts note the \$1.5 million price tag, plus annual maintenance costs, puts the system out of reach for most community hospitals.

Intuitive Surgical expects installations in more than 100 hospitals by end of 2001.

Surgeon in New York Operates on Patient in France — Without Leaving His Chair

NEW YORK — In a procedure that surgeons are calling the “Lindbergh Operation,” a team led by Dr. Jacques Marescaux removed a gallbladder from a 68-year-old patient in Strasbourg, France, while Dr. Marescaux sat at a robotic control console in Manhattan. The operation took 54 minutes. The patient was discharged two days later with no complications.

The surgery was made possible by a dedicated high-speed fiber optic cable leased between New York and Strasbourg, combined with a Zeus robotic surgical system capable of translating the surgeon’s hand movements into instrument actions with a signal delay of under 200 milliseconds, fast enough, the team argues, to be safe.

“What we proved is that distance is no longer the barrier we thought it was,” Dr. Marescaux said in a statement. “The Lindbergh flight showed the Atlantic could be crossed. This shows surgery can cross it too.”

Not everyone shares the enthusiasm. Dr. Patricia Wells, chair of surgery at Columbia University Medical Center, urged caution about reading too much into a single case. “One successful procedure over a purpose-built private cable is a long way from a clinical reality,” she said. “The infrastructure required to do this reliably, at scale, in anything other than ideal conditions doesn’t exist.” Others noted that the cost of the dedicated fiber link (approximately one million dollars) made the demonstration more symbolic than practical for now.

Still, observers across the surgical community agreed on one point: the operation proved that telesurgery is no longer science fiction. What remains unclear is how long it takes to become medicine.

Robotic System Used to Remove 60% of a Patient's Liver — in Four Incisions

CHICAGO — A team at the University of Illinois Hospital has performed a successful robotic-assisted right hepatectomy, removing more than sixty percent of a patient's liver, using the da Vinci Surgical System. The procedure, used to treat a primary liver tumour, was completed through four incisions each less than a centimetre wide. The patient was discharged on the fourth day following surgery; the same procedure performed as open surgery typically requires a hospital stay of seven to ten days.

"This changes the ceiling of what robotic surgery can achieve," said the lead surgeon. "Even the most complex abdominal surgery is within reach of this technology now."

The result is significant because liver surgery has historically been considered too high-risk for minimally invasive approaches, due to the organ's fragility and vascularity. Its success will likely accelerate adoption of robotic platforms beyond urology and gynaecology, where they have been concentrated until now.

Critics were quick to note the gap between a landmark case and routine practice. "One exceptional team achieving this in an academic centre is very different from it being safe and reproducible across the surgical community," said a hepatobiliary surgeon at Toronto General. "The learning curve for robotic liver surgery is steep. We need outcomes data from many centres, not one impressive result." Program directors also raised concerns about training pipelines, noting that access to robotic systems for residents remains inconsistent and that credentialing standards vary widely between institutions.

The Surgeon of the Future Might Never Enter the Operating Room

SAN FRANCISCO — The Lindbergh Operation happened three years ago, and most of medicine has moved on. Researchers who worked on it haven't.

At laboratories in MIT, Stanford, and several European universities, a quiet race is underway to solve the three remaining barriers to viable telesurgery: latency, haptic feedback, and infrastructure cost. Latency (the signal delay between a surgeon's movement and the instrument's response) is the most critical. Current fiber optic networks can achieve delays under 150 milliseconds on dedicated lines. Engineers believe 50 milliseconds is the safe threshold for real-time surgical control. Closing that gap is, at current rates of infrastructure investment, a matter of years, not decades.

Haptic feedback is the subtler problem. Current robotic systems like da Vinci offer no tactile information: the surgeon sees but cannot feel. Research teams are developing glove-based interfaces that simulate tissue resistance and texture in real time. "When haptics work, the last serious objection to robotic surgery collapses," said one MIT researcher.

There are those who remain skeptical that the broader vision will ever land. "We keep hearing that telesurgery is five years away," said Dr. Harold Fenwick, a medical ethicist at the University of Toronto. "It has been five years away for five years. The technical pieces matter less than the regulatory, legal, and liability frameworks, and nobody is building those." Others point to the surgeon workforce gap in rural and low-income regions as the real argument for urgency: Sub-Saharan Africa has one surgeon per 100,000 people. Canada has twenty-three. If telesurgery scales, that equation changes.

Surgical Simulation: Are Video Game Controls the Future of Training?

Thread started March 14, 2001 · 5 Replies · 1,204 Views

OP **futuremed_kira** Tue Mar 14, 2001 9:22 am

So I got to try one of those new surgical simulators last month during a lab visit. It's basically a video game controller hooked up to a box that pushes back when you press on virtual tissue. You practice laparoscopic techniques on a screen before you ever touch a real patient.

Honest reaction: it felt like a video game. And I mean that in a good way. I'm a decent gamer. I picked up the motions fast. But my attending (who's been doing this for 20 years) struggled more than I did at first. That felt significant to me.

I know anecdotes aren't data. But is this the direction things are going? Are hospitals going to start caring whether you grew up with a Nintendo?

— kira · MS3 @UofT Medicine · "The map is not the territory"

REPLY **scalpel_skeptic** Tue Mar 14, 2001 11:47 am

Surgery is not a video game. Never will be. The things a simulator cannot teach you: bleeding that doesn't stop on schedule, tissue that doesn't behave like the model, a patient who moved. The unpredictability is the whole point.

I don't doubt that simulators have some value for basic motor skill drills. But the idea that being good at Zelda makes you a better surgeon is the kind of thing that sounds smart in a coffee shop and falls apart in an OR. Clinical judgment comes from experience with real patients, not from racking up simulator scores.

— "A good surgeon knows how to operate; a great surgeon knows when not to."

REPLY **futuremed_kira** Tue Mar 14, 2001 1:15 pm

Fair pushback. I'm not saying it replaces real experience. I'm saying it might reduce the number of real-patient mistakes it takes to get there.

Flight simulators don't replace flying. But nobody argues airline pilots shouldn't use them because "real turbulence is unpredictable." The simulator handles the motor memory so the actual flight doesn't have to. Seems like the same logic should apply here.

— kira · MS3 @UofT Medicine · "The map is not the territory"

REPLY ResidentRealTalk Wed Mar 15, 2001 8:34 am

The flight analogy is everywhere these days. I like it. But there's a gap: aviation has had standardized simulators integrated into training for 30+ years. Surgery has had them for what, three? The infrastructure for this to matter (standardized curricula, validation studies, hospital buy-in) is at least 15 years away from maturity. Maybe more.

I'm not saying don't develop them. I'm saying let's not get ahead of what the evidence currently supports.

— PGY-2 General Surgery · *"Evidence first, enthusiasm second"*

REPLY laparolearner99 Wed Mar 15, 2001 11:19 am

Actually there is some early data. Vanderbilt ran a study where surgical residents who played video games more than 3 hours per week made 37% fewer errors on a laparoscopic simulator. And they completed tasks 27% faster. It's a small study. But it wasn't random noise.

Nobody's saying video games = surgery. The fine motor mapping is the interesting signal. And I think kira's point stands: we might want to stop treating gaming as irrelevant to manual dexterity training.

— *"Hands first. Everything else second."*

DARPA Built This Technology — But Who Pays for the Cable?

Thread started January 29, 2004 · 4 Replies · 4,812 Views

OP **bandwidth_hawk** *Thu Jan 29, 2004 7:02 pm*

People forget this technology didn't come out of nowhere. DARPA funded the original remote surgery research in the 1990s specifically for battlefield use: operating on wounded soldiers too dangerous to move, via robot, from a safe location. The Lindbergh Operation in 2001 was essentially a civilian demonstration of military technology that had been in development for a decade.

The question now isn't can it work; we know it can. The question is who builds the network infrastructure that makes it work at scale, outside of a one-off \$1M private fiber cable. The technology leads. The economics lag. This is always how it goes.

— *"The network is the computer."* (Sun Microsystems, 1984)

REPLY **pragmatic_MD** *Thu Jan 29, 2004 9:45 pm*

The technology concern isn't the cable. It's the latency. The Lindbergh team got below 200 milliseconds over that private line. But surgeons need feedback in under 200ms to work safely; 300ms feels "wrong" and could cause errors. On commercial internet infrastructure, you regularly see 150–400ms depending on load. The variance is the problem, not the average.

You can't have a surgeon in the middle of a critical maneuver and then have a half-second delay spike because someone in their ISP's area is downloading a Linux distribution. This isn't a solvable problem by making the cable better. It's a network architecture problem.

— *Hospitalist. "Perfect is the enemy of good. Except in surgery."*

REPLY **bandwidth_hawk** *Fri Jan 30, 2004 8:11 am*

That's fair but the Lindbergh team specifically measured and sub-200ms is achievable. The issue is consistency. You need a dedicated QoS channel, not best-effort internet. That's an engineering problem, not a physics problem. Dedicated surgical network channels with guaranteed latency bands are a solved problem conceptually; it's the same idea as voice-over-IP QoS. The deployment cost is real, but it's not insurmountable.

— *"The network is the computer."* (Sun Microsystems, 1984)

REPLY **OR_Realist** *Fri Jan 30, 2004 11:53 am*

"Deployment cost is real but not insurmountable" — this phrase is doing a lot of work. The private fiber cable for the Lindbergh case cost \$1 million. For one surgery. Who is paying for that infrastructure at scale? Provincial health budgets? Private insurers? Hospital systems already running on 2% margins? The technology works in a lab. The economics don't work in a health system.

I'll believe this is a clinical pathway when someone shows me the billing code and the reimbursement model. Until then, it's a very expensive proof of concept.

— *Hospital administrator. "Follow the money, then follow the care."*

REPLY **NorthernOntarioDoc** *Fri Jan 30, 2004 4:27 pm*

Posting from northern Ontario where we have 1 general surgeon for about 80,000 people. My patients routinely drive 4+ hours to Toronto for procedures that should be available locally. The economics argument is real, but I'd ask OR_Realist to consider the current cost of NOT having this technology. We're already paying for patient transfers, lost work, and outcomes that degrade during transport.

I'm not saying telesurgery is ready. I'm saying the people calling it impractical tend to practice in cities where surgical access is not a problem. For us, impractical is the current system.

— *GP/FP, Sudbury Regional · "Distance is a diagnosis up here."*

Is the da Vinci System Worth \$1.5 Million? Or Is This Just Marketing?

Thread started September 5, 2002 · 4 Replies · 3,109 Views

OP **budget_blade** *Fri Sep 5, 2002 10:14 am*

Hospital CFO perspective. We've been pitched the da Vinci system twice in the past year. The sales deck is impressive. The peer-reviewed evidence is... thin. There's no randomized controlled trial showing da Vinci outperforms skilled conventional laparoscopy on outcomes that matter to patients: mortality, complication rate, readmission.

What I see: hospitals buying a \$1.5M robot because their competitors across town bought one and they don't want to lose market share. That's a marketing-driven arms race, not evidence-based purchasing. I'm not saying the technology doesn't work. I'm saying we're paying for something we can't yet prove is better.

— *"If you can't measure it, you can't manage it."*

REPLY **intuitive_investor** *Fri Sep 5, 2002 2:38 pm*

The comparison baseline matters. You're comparing da Vinci to "skilled conventional laparoscopy," which takes years to develop and has its own steep learning curve. Compare da Vinci to open surgery, which is the realistic alternative for most procedures, and the picture changes: smaller incisions, less blood loss, shorter hospital stays. Those translate directly to patient outcomes and cost savings that offset the capital expenditure over 5–10 years.

Also: the RCT ship has sailed. You're not going to get informed consent for a trial where half the patients get a worse surgical approach. The evidence base will be observational data from here.

— *Long ISRG. This is not investment advice.*

REPLY **budget_blade** *Sat Sep 6, 2002 9:01 am*

I'm not going to get investment advice from a self-disclosed ISRG long. But the point about comparison baseline is fair. Open surgery is a higher bar to clear and da Vinci clears it. My concern is the way it's being marketed. I've seen hospital billboards advertising "robotic surgery" as if the robot is the surgeon. Patients are requesting it specifically because of the marketing. That's not how we should be making surgical decisions.

— *"If you can't measure it, you can't manage it."*

REPLY **MedEd_Observer** *Sat Sep 6, 2002 1:44 pm*

The conversation about cost and evidence is right. But nobody is talking about the training bottleneck. I work in medical education. Residents currently learn to use the da Vinci system by... watching, mostly. OR time is limited. The machine is expensive so it gets reserved for revenue-generating procedures. Trainees get minimal hands-on time before being expected to use it clinically.

The question isn't just whether the robot is worth the money. It's whether we're investing in training infrastructure at the same pace as the hardware. Right now we are not.

— *Medical education research, Queen's · "We teach the way we were taught."*

REPLY **FutureOR_2020** *Sun Sep 7, 2002 10:22 am*

Training gap is real. But I'd bet every hospital system that hasn't bought a da Vinci by 2005 is going to be playing catch-up for a decade. The adoption curve on this is going to look like the EMR curve: skeptics now, mandates later.

And budget_blade's concern about the marketing will sort itself out. Once you have enough outcome data from real centres, the clinical case either stands or doesn't. I think it will. Feel free to find this post in 2010 and tell me I was wrong.

— *"Bet on the direction of travel, not the current stop."*

Why Video Game Kids Will Be Better Surgeons Than Me

Posted by **Dr. A. Reeves** · FutureMed.blogspot.com · February 14, 2003

I am 41. I did not grow up with video games. I learned laparoscopic surgery the old-fashioned way: watch one, do one, teach one. It took years. There were bad months early on.

Last week I watched a third-year resident use our new surgical simulator, the kind with a haptic controller that pushes back when you press on virtual tissue. She had never touched a real laparoscope. She moved through the interface like she'd been using it for years. Smooth, controlled, spatially confident. I asked her how much gaming she'd done growing up.

"A lot," she said.

There's now a peer-reviewed study from Vanderbilt (33 participants, not a huge sample) showing that surgeons who played video games more than 3 hours per week made 37% fewer errors on a laparoscopic simulator and were 27% faster. The effect was particularly strong for the tool-manipulation tasks that are genuinely hard to learn on a live patient without causing harm first.

I'm not saying we should start screening surgical applicants based on their Call of Duty stats. But I am saying: the generation coming up behind me is going to have a different baseline than I did. And that probably matters.

The more interesting question for me is what this means about simulation-based training. If fine motor skills developed in gaming transfer to surgical tools, can we build more of that pipeline systematically, before residents ever step into an OR?

4 COMMENTS

DrOldSchool · February 14, 2003 · 11:32 am

33 people is not a study, it's a conversation. You cannot build a training paradigm on n=33. I've been in surgical education for 25 years. The things this study can't measure are the things that kill patients: judgment under pressure, knowing when to call for help, recognizing the limit of your own skill. You can't simulate those.

futuremed_kira · February 14, 2003 · 2:47 pm

The claim being made is modest: video game experience correlates with simulator performance on motor tasks. Not that it replaces clinical judgment. Dr. Reeves isn't saying gamers make better surgeons; they're saying motor skill transfers. That's a narrower and more defensible claim than DrOldSchool is arguing against.

SimSurgery_Dev · February 15, 2003 · 9:04 am

We build surgical simulators. The data we're seeing internally is consistent with the Vanderbilt findings, especially on tool-tip control and bimanual coordination. The 37% error reduction number is striking but not surprising. The more interesting finding is that simulator-trained residents seem to reach proficiency on OR tasks faster. That's the variable that actually matters for training programs.

DrOldSchool · February 15, 2003 · 4:18 pm

Fair. If the claim is specifically about motor skill transfer and not clinical judgment, I'm less resistant to it. My concern is that "video game kids will be better surgeons" is a headline that collapses that distinction in ways that could distort how we train and select residents. Motor dexterity is necessary. It is not sufficient. I'd hate to see programs screen out people with great instincts and poor gaming backgrounds.

Haptic Feedback: The Missing Piece in Robotic Surgery

Posted by **K. Bauer** · IEEE Spectrum.org · March 8, 2004

If you've watched a da Vinci procedure, you've probably noticed that the surgeon's hands are doing something strange: they're working in mid-air, without touching anything. That's because the current generation of robotic surgical systems provides no haptic feedback. The surgeon can see with superhuman clarity. They cannot feel at all.

This is not a minor limitation. Touch is how surgeons know how hard to push. It's how they distinguish healthy tissue from diseased tissue when the visual cues are ambiguous. Experienced surgeons have spent years developing tactile intuition about what's normal and what isn't. Robotic surgery strips all of that out and replaces it with visual cues alone.

The engineering challenge of haptic feedback in robotic surgery is genuinely hard. You need sensors at the instrument tip that can detect tissue resistance, shear force, and vibration frequency, then transmit that information in real time to a glove or controller at the surgeon's hand, all with latency under 5 milliseconds or the brain will perceive it as artificial. We are close to crossing that threshold. Not there yet, but close.

When we get there, the last serious objection to robotic surgery as a full replacement for laparoscopic approaches will have been addressed.

3 COMMENTS

SurgeonNotRobot · March 8, 2004 · 6:11 pm

"We are close." How many times have I read this sentence about haptic feedback in surgery? I've been reading it since 1998. It has been the next breakthrough for six years straight. At what point do we call it what it is: a perpetually-five-years-away technology?

K_Bauer_Author · March 9, 2004 · 10:22 am

That's a fair critique. "Close" has been misused in this space. What I'd say is: we've crossed some specific thresholds in the last 18 months that weren't crossed in 1998: sub-10ms latency in lab conditions, miniaturized sensor arrays that fit through a 5mm port. The problem feels different now than it did then. I might be wrong. But I don't think I'm repeating old promises.

haptics_researcher · March 9, 2004 · 3:44 pm

The hard number is the $<5\text{ms}$ threshold you mention. Most lab demonstrations right now are operating at 8–12ms, which is perceptible as artificial. Getting below 5ms in a clinical environment (not a clean lab) is the genuine remaining challenge. Sensor noise, signal conditioning, the physical length of cables through the instrument arm: these all add up. Achievable? Yes. Imminent? That depends on your funding timeline.

Robots in the OR: Why I'm Not Worried

Posted by **Dr. W. Hartley, retired general surgeon** · SkepticalScalpel.com · August 22, 2003

Every few weeks someone asks me whether I'm worried about robots taking surgeons' jobs. My answer is no. Here's why.

Surgery is a craft, not a protocol. The moment in an operation that determines success is rarely the planned moment. It's the unexpected finding, the aberrant anatomy, the bleeder in the wrong place. Those moments require judgment, adaptation, and experience in ways that no current robotic system can remotely approximate. The machine that can be taught to follow a procedure cannot be taught to handle the procedure going wrong.

The analogy to radiology doesn't hold. Yes, AI is beginning to read scans with radiologist-level accuracy on defined tasks. But radiology, at its most automatable edge, is fundamentally about pattern recognition in a static image. Surgery is about navigating three-dimensional biological variability in real time, with a live patient whose physiology is changing under your hands. These are categorically different problems.

I'm not anti-robot. The da Vinci is a useful tool in skilled hands. But a tool is not a replacement. The surgeon is still the surgeon. The robot is still the instrument.

4 COMMENTS

futuremed_kira · August 22, 2003 · 9:18 pm

The radiology comparison is interesting because radiologists said something similar 10 years ago, that their work required too much judgment to automate. That's still probably true at the expert level. But at the threshold-screening level, AI is already performing peer with junior radiologists on specific tasks like nodule detection. The question isn't whether surgery is harder to automate than radiology. It is. The question is whether that means no automation ever, or just slower automation.

Dr_Hartley_Author · August 23, 2003 · 8:04 am

Slower automation and no automation are meaningfully different. If it takes 40 years to automate 20% of surgical tasks, that's not "surgery is being replaced," that's "surgery is evolving." I'm arguing against the implication that we should worry now. Not that zero change is coming.

DARPA_watcher · *August 23, 2003 · 11:30 am*

Worth noting: DARPA's current research goal for surgical robots is semi-autonomous operation for specific subtasks (suturing, stapling, tissue retraction), not full replacement. Semi-autonomous is a very different threat model than full automation and should be treated differently in this conversation.

Dr_Hartley_Author · *August 23, 2003 · 2:55 pm*

DARPA_watcher makes a good point. Semi-autonomous subtask execution is something I'd welcome; it's essentially a better instrument. Fully autonomous surgery is the premise I was arguing against. If the realistic trajectory is the former, most of my concern dissolves.

Why Patients Will Never Fully Accept a Robot Operating on Them

Posted by **Prof. C. Alderton, University of Edinburgh** · MedEthics.org · Bioethics Quarterly · November 3, 2004

There is a consent problem at the heart of robotic surgery that almost nobody is discussing: most patients, when surveyed, say they do not want a robot operating on them. They want a human surgeon. They will accept a surgeon using a robotic tool. They will not accept the robot as the primary agent.

This is not irrationality. It is a sensible response to a genuine ethical question: who is responsible when something goes wrong? With a human surgeon, the answer is clear. With a semi-autonomous robotic system, even one nominally controlled by a surgeon, the chain of accountability becomes murky. Is it the surgeon? The hospital? The manufacturer? The software team that programmed the motion-planning algorithm?

As robotic surgery systems become more autonomous (and they will, incrementally) this question of accountability becomes more acute, not less. We are building the technology faster than we are building the ethical and legal frameworks to govern it. The public's intuitive unease about robotic surgery is not ignorance. It is a response to a real gap.

3 COMMENTS

RoboSurgAdvocate · November 3, 2004 · 7:44 pm

Current robotic surgical systems are not semi-autonomous in any meaningful sense. The da Vinci does nothing unless a surgeon actively controls it. The surgeon is the primary agent; the robot is the instrument. The accountability question you're raising applies to the future, not to present clinical practice. Framing it as a current consent problem is premature.

Prof_Alderton_Author · November 4, 2004 · 9:15 am

You're right that current systems are surgeon-controlled. My concern is directional: the research trajectory is toward greater autonomy: autonomous suturing, autonomous tissue retraction. The ethical frameworks need to be built now, before the capability arrives, not after. That's the argument. Not that today's da Vinci procedure requires a new consent form.

PatientAdvocate_MB · November 4, 2004 · 1:38 pm

The survey data on patient acceptance may be more mixed than implied. I've seen patients in rural settings who are very willing to accept robotic or telesurgical approaches precisely because the alternative is a 6-hour drive and a 3-month wait. Acceptance may depend heavily on what the patient's actual alternative is. The "patients won't accept it" framing may reflect urban, adequately-served patients more than the full picture.

T1

@MarcescauxSurg, Jacques Marescaux · Sep 8, 2001

We just completed the first transatlantic surgery. Patient in Strasbourg, surgeon in New York. Gallbladder removed via robotic arm over fiber optic cable. The "Lindbergh Operation." Surgery without borders is no longer science fiction.

🔄 312 re-posts · ❤️ 891 likes

T2

@DrSkeptic_OR · Sep 9, 2001

↩ Replying to @MarcescauxSurg

One procedure on a \$1M dedicated fiber cable, with a full DARPA team standing by, is not a clinical pathway. Congrats on the milestone, but let's not pretend this is replicable on any hospital's budget or any country's internet infrastructure.

🔄 44 re-posts · ❤️ 203 likes

T3

@SurgTechConf, SurgTech Conference · Nov 3, 2004

Broadband now in 50% of North American homes. Surgeons at today's panel asking: what happens to telesurgery latency when we hit gigabit speeds everywhere? The answer might change everything about who can operate and from where. #SurgTech2004

🔄 88 re-posts · ❤️ 204 likes

T4

@DrRuralNorth, RuralMD Northern Ontario Feb 19, 2005

↩ Replying to @SurgTechConf

People keep asking me about "telesurgery." My hospital doesn't have reliable broadband. I can barely video call a specialist in Edmonton. The idea that I'll be operating remotely in my lifetime is science fiction. Focus on the basics please.

📩 29 re-posts · ❤ 117 likes

T5

@GHWatch, GlobalHealth Watch Aug 22, 2005

Sub-Saharan Africa: 1 surgeon per 100,000 people. Canada: 23. If robotic telesurgery scales, a surgeon in Toronto could treat a patient in Nairobi. The equity implications are enormous and almost nobody in this conversation is talking about them. #GlobalSurgery

📩 54 re-posts · ❤ 143 likes

T6

@MedResWeekly Oct 1, 2004

AR overlay in surgery: Stanford team projecting CT scan data directly onto the patient's body during live operation. Surgeon sees tumour location through the skin in real time. Still experimental. But this is unmistakably the direction.

#AugmentedReality #Surgery

📩 167 re-posts · ❤ 441 likes

T7**@ORNurse_Ottawa** Oct 3, 2004[↩ Replying to @MedResWeekly](#)

"Unmistakably the direction" is what people said about voice-activated surgery in 1998. About telemedicine in 2000. Every promising OR technology takes 20 years longer than the excited researcher predicted. I'll believe it when we're training it in residency programs.

 31 re-posts ·  289 likes**T8****@ResidentRealTalk** May 12, 2005

Reality check: the da Vinci at our hospital sits idle 60% of the time because there aren't enough trained operators. Meanwhile residents can't get time on it because it's being reserved for attending-level procedures. The training gap is going to limit adoption just as much as cost.

 201 re-posts ·  876 likes**T9****@SimSurgery_Dev** Jun 3, 2005[↩ Replying to @ResidentRealTalk](#)

This is exactly why simulation-based training matters. If residents can build proficiency on a simulator before they touch the actual robot, OR access time requirements drop significantly. We have data showing simulation-trained residents reach OR competency with 40% fewer supervised live cases. The hardware isn't the bottleneck; the training pipeline is.

 89 re-posts ·  334 likes

T10

@DARPA_MedWatch Jul 19, 2005

DARPA's Trauma Pod program is targeting semi-autonomous surgical robots for battlefield use: automated suturing, wound management, hemorrhage control without a surgeon present. The stated goal is a system deployable by 2010. This is where the civilian OR technology is going to come from next. #DARPA #FutureOfSurgery

143 re-posts · 502 likes

A1

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A2

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A3

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